7.2 Amazon S3

Amazon S3 is a distributed data store, and one of the oldest services provided by AWS. *Amazon S3* is an acronym for *Amazon Simple Storage Service*. It's a typical web service that lets you store and retrieve data organized as objects via an API reachable over HTTPS. Here are some typical use cases:

- *Storing and delivering static website content* —For example, our blog https://cloudonaut.io is hosted on S3.
- *Backing up data*—For example, you can back up your photo library from your computer to S3 using the AWS CLI.
- *Storing structured data for analytics, also called a data lake*—For example, you can use S3 to store JSON files containing the results of performance benchmarks.
- *Storing and delivering user-generated content*—For example, we built a web application—with the help of the AWS SDK—that stores user uploads on S3.

Amazon S3 offers virtually unlimited storage space and stores your data in a highly available and durable way. You can store any kind of data, such as images, documents, and binaries, as long as the size of a single object doesn't exceed 5 TB. You have to pay for every GB you store in S3, and you also incur costs for every request and for all transferred data. As figure 7.2 shows, you can access S3 via the internet using HTTPS to upload and download objects. To access S3, you can use the Management Console, the CLI, SDKs, or third-party tools.



SFigure 7.2 Uploading and downloading an object to S3 via HTTPS

3 uses *buckets* to group objects. A bucket is a container for objects. It is up to you to create multiple buckets, each of which has a globally unique name, to separate data for different scenarios. By *unique*, we really mean unique—you have to